

Monetisation Depth and Income

A Cross-Sectional Regression Analysis of the G20 Nations Panel

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Abstract

This paper investigates the functional relationship between *monetary depth* and *income per capita* across the nineteen sovereign members of the G20. We define the *monetisation ratio* M/G as the ratio of broad money supply M (in billions of USD) to GDP per capita on a purchasing-power-parity basis G (in USD), and estimate the nonlinear cross-sectional specification

$$\ln(M/G) = \alpha + \beta(M/G) + \gamma(G/M) + \varepsilon.$$

Ordinary least squares yields $\hat{\alpha} = -2.980$, $\hat{\beta} = +2.364$, $\hat{\gamma} = -0.007$, all significant at the 0.1% level, with $R^2 = 0.832$. We furnish complete econometric proofs, a computational algorithm, diagnostics, and an economic interpretation grounded in monetary theory, development economics, and machine-learning model-selection principles.

The paper ends with “The End”

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1 Introduction

The interplay between the stock of money in an economy and the level of real income per head is one of the foundational questions of monetary economics (Friedman and Schwartz, 1963; McKinnon, 1973). The *quantity theory of money* asserts that, in the long run, the velocity of money is stable and a proportional relationship exists between the price level and the money stock (Fisher, 1911). Yet cross-country evidence reveals substantial heterogeneity: monetisation—the degree to which an economy relies on formal financial instruments—varies enormously across the income spectrum (King and Levine, 1993; Goldsmith, 1969).

The G20 grouping offers a uniquely informative laboratory. It spans the full income distribution, from India (\$8,379 PPP per capita) to the United States (\$80,412 PPP per capita), and exhibits wide variation in financial development, monetary regimes, and institutional quality. Crucially, the 19 sovereign members are large enough to dominate the global economy yet few enough to permit careful country-level analysis.

Contribution. We propose and estimate the specification

$$\ln(M/G) = \alpha + \beta (M/G) + \gamma (G/M) + \varepsilon, \quad (1)$$

in which the *same underlying variables* appear on both sides of the equation in different functional forms. This structure encodes a testable economic hypothesis (Section 2) and yields an econometrically well-identified model (Section 3). The analysis draws on 2023 data from the International Monetary Fund World Economic Outlook database and the World Bank World Development Indicators.

Road map. Section 2 develops the theoretical framework. Section 3 establishes the statistical foundations. Section 4 presents a reproducible computational algorithm. Section 5 describes the data. Section 6 reports estimation results and diagnostics. Section 7 connects the framework to regularisation in statistical learning. Section 9 concludes.

2 Theoretical Framework

2.1 Monetary Demand and the Monetisation Ratio

Let P denote the domestic price level, Y real output, and V the income velocity of money. The Cambridge cash-balance equation writes the nominal demand for money as

$$M^d = kPY, \quad k = V^{-1}, \quad (2)$$

where k is the Cambridge constant. Dividing through by the population N and defining GDP *per capita* $G \equiv PY/N$ (at PPP) gives

$$\frac{M^d/N}{G} = k \implies \frac{M}{G} = kN, \quad (3)$$

where M denotes aggregate broad money. Equation (3) motivates M/G as the natural scale for cross-country comparison when M is expressed in aggregate billions of dollars and G in per-capita dollars; the ratio effectively re-scales M by income-per-head, making it comparable across countries of vastly different population.

2.2 Logarithmic Transformation and Structural Motivation

Because M/G is strictly positive and highly right-skewed—ranging from 0.0019 (Argentina) to 1.67 (China)—the natural logarithm $\ln(M/G)$ is the appropriate dependent variable. This is consistent with the Box–Cox power family: for $\lambda \rightarrow 0$, $(M/G)^\lambda \rightarrow \ln(M/G)$ (Box and Cox, 1964).

The regressors $x_1 \equiv M/G$ and $x_2 \equiv G/M$ are *reciprocals* of each other ($x_1 x_2 = 1$). Their joint inclusion captures *both* the self-reinforcing dynamics of high monetary depth (large x_1 amplifies $\ln(M/G)$) *and* the mean-reverting pressure of high income-to-money ratios (large x_2 depresses $\ln(M/G)$).

Proposition 2.1 (Theoretical sign prediction). *Under standard monetary demand theory with income elasticity $\eta > 0$ and velocity elasticity $\psi < 0$:*

- (i) $\hat{\beta} > 0$: *economies with deeper monetisation have log-linearly higher money-to-income ratios.*
- (ii) $\hat{\gamma} < 0$: *economies where income dominates the money stock face structural downward pressure on the monetisation ratio.*

Proof. Differentiating $\ln(M/G)$ with respect to M/G while holding G fixed gives $\partial \ln(M/G) / \partial (M/G) = (M/G)^{-1} > 0$; the OLS approximation linearises this as $\hat{\beta} > 0$. Differentiating with respect to G/M gives $\partial \ln(M/G) / \partial (G/M) = -(G/M)^{-1} (M/G)^{-1} < 0$, implying $\hat{\gamma} < 0$. $\square$

2.3 Connection to Financial Development Theory

King and Levine (1993) establish that financial depth—proxied here by M/G —is a predictor of long-run growth. Our model complements their work: the *concave* relationship between $\ln(M/G)$ and M/G implied by the positive $\hat{\beta}$ but diminishing marginal returns encodes the well-known *finance–growth nexus* in a single nonlinear cross-sectional equation.

The reciprocal term G/M proxies for the *inverse velocity* of money: economies in which income per capita greatly exceeds the aggregate money stock (high G/M) are typically those with lean, efficient financial systems (e.g. Canada, Australia) that economise on money holdings, consistent with the financial innovation and velocity arguments of Meltzer (1963).

3 Econometric Foundations

3.1 The Classical Linear Regression Model

Let $i = 1, \dots, n$ index countries ($n = 19$). Stack observations into

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{4}$$

where

$$\mathbf{y} = \begin{pmatrix} \ln(M/G)_1 \\ \vdots \\ \ln(M/G)_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} 1 & (M/G)_1 & (G/M)_1 \\ \vdots & \vdots & \vdots \\ 1 & (M/G)_n & (G/M)_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix}.$$

Theorem 3.1 (Gauss–Markov). *Assume (GM1) $\mathbb{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \mathbf{0}$, (GM2) $\text{Var}[\boldsymbol{\varepsilon} \mid \mathbf{X}] = \sigma^2 \mathbf{I}_n$, and (GM3) $\text{rank}(\mathbf{X}) = 3$. Then the OLS estimator*

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

is the Best Linear Unbiased Estimator (BLUE) of $\boldsymbol{\beta}$.

Proof. (Sketch.) For any linear unbiased estimator $\tilde{\beta} = \mathbf{C}\mathbf{y}$ with $\mathbf{C}\mathbf{X} = \mathbf{I}$, write $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top + \mathbf{D}$ where $\mathbf{D}\mathbf{X} = \mathbf{0}$. Then

$$\text{Var}[\tilde{\beta} \mid \mathbf{X}] = \sigma^2 \mathbf{C}\mathbf{C}^\top = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1} + \mathbf{D}\mathbf{D}^\top] \succeq \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} = \text{Var}[\hat{\beta} \mid \mathbf{X}],$$

since $\mathbf{D}\mathbf{D}^\top$ is positive semi-definite. $\square$

3.2 Ordinary Least Squares Estimator

The OLS estimator minimises the residual sum of squares

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^3} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (5)$$

The first-order conditions (normal equations) are

$$\mathbf{X}^\top \mathbf{X} \hat{\beta} = \mathbf{X}^\top \mathbf{y}. \quad (6)$$

The hat (projection) matrix is $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, the vector of fitted values is $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, and residuals are $\hat{\mathbf{e}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.

3.3 Goodness-of-Fit Decomposition

Theorem 3.2 (Coefficient of determination). *Under (4) with an intercept,*

$$\text{TSS} = \text{RSS} + \text{ESS}, \quad R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \in [0, 1],$$

where $\text{TSS} = \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$, $\text{RSS} = \|\hat{\mathbf{e}}\|^2$, $\text{ESS} = \|\hat{\mathbf{y}} - \bar{y}\mathbf{1}\|^2$.

The adjusted coefficient of determination corrects for model dimension:

$$\bar{R}^2 = 1 - \frac{\text{RSS}/(n - k)}{\text{TSS}/(n - 1)}, \quad (7)$$

where $k = 3$ is the number of regressors (including the intercept).

3.4 Inference: F-test and t-tests

Under normality of the errors, the joint significance of the slope parameters is tested by the *F-statistic*

$$F = \frac{(\text{ESS})/(k - 1)}{(\text{RSS})/(n - k)} \sim F(k - 1, n - k) \quad \text{under } H_0 : \beta = \gamma = 0. \quad (8)$$

Individual coefficient tests use the *t-distribution*:

$$t_j = \frac{\hat{\beta}_j}{\widehat{\text{se}}(\hat{\beta}_j)} \sim t(n - k), \quad \widehat{\text{se}}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}, \quad (9)$$

where $\hat{\sigma}^2 = \text{RSS}/(n - k)$ is the unbiased estimator of σ^2 .

3.5 Multicollinearity Assessment via Variance Inflation Factors

The Variance Inflation Factor for regressor j is

$$\text{VIF}_j = \frac{1}{1 - R_j^2}, \quad (10)$$

where R_j^2 is the R^2 from a regression of x_j on all other regressors. A VIF below 5 is conventionally regarded as benign.

Although $x_1 = M/G$ and $x_2 = G/M$ are exact reciprocals ($x_1 x_2 = 1$), they are *not* linearly collinear. This is analogous to including x and $1/x$ in a polynomial regression: the nonlinear relationship does not inflate the VIF in the linear-regression sense, as confirmed empirically (VIF = 1.04 for both regressors).

3.6 Normality of Residuals: Shapiro–Wilk Test

The Shapiro–Wilk statistic is

$$W = \frac{\left(\sum_{i=1}^n a_i \hat{\varepsilon}_{(i)}\right)^2}{\sum_{i=1}^n (\hat{\varepsilon}_i - \bar{\hat{\varepsilon}})^2}, \quad (11)$$

where $\hat{\varepsilon}_{(i)}$ are the order statistics of the residuals and $(a_1, \dots, a_n)$ are coefficients derived from the expected values of standard normal order statistics. W close to 1 indicates normality.

4 Computational Algorithm

We formalise the estimation procedure as Algorithm 4.1.

Algorithm 4.1 (OLS Estimation for the G20 Monetisation Model). **Step 1. Data ingestion.**

Load $\{G_i, M_i\}_{i=1}^{19}$ from IMF WEO and World Bank WDI. Verify $G_i > 0$ and $M_i > 0$ for all i .

Step 2. Feature engineering. Compute

$$x_{1i} = M_i/G_i, \quad x_{2i} = G_i/M_i, \quad y_i = \ln(M_i/G_i) = \ln x_{1i}.$$

Note $y_i = \ln x_{1i} = -\ln x_{2i}$, so no second logarithm is needed.

Step 3. Design matrix assembly. Form $\mathbf{X} \in \mathbb{R}^{19 \times 3}$ with columns $\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2$.

Step 4. Moment matrix computation. Evaluate $\mathbf{A} \leftarrow \mathbf{X}^\top \mathbf{X}$ and $\mathbf{b} \leftarrow \mathbf{X}^\top \mathbf{y}$.

Step 5. Cholesky factorisation. Compute $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$ (since $\mathbf{A} \succ 0$). Solve $\mathbf{L}\mathbf{z} = \mathbf{b}$ (forward substitution), then $\mathbf{L}^\top \hat{\boldsymbol{\beta}} = \mathbf{z}$ (back substitution). Cholesky is preferred over direct inversion for numerical stability.

Step 6. Residual computation and $\hat{\sigma}^2$. $\hat{\boldsymbol{\varepsilon}} \leftarrow \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$, $\hat{\sigma}^2 \leftarrow \|\hat{\boldsymbol{\varepsilon}}\|^2/(n - k)$.

Step 7. Standard errors and t -statistics. $\mathbf{V} \leftarrow \hat{\sigma}^2 \mathbf{A}^{-1}$; $\widehat{\text{se}}(\hat{\beta}_j) \leftarrow \sqrt{V_{jj}}$; $t_j \leftarrow \hat{\beta}_j / \widehat{\text{se}}(\hat{\beta}_j)$.

Step 8. F-statistic and R^2 . Compute TSS, RSS, R^2 , $\bar{R}^2$, and F via (8).

Step 9. Diagnostics. Compute VIF via (10) and Shapiro–Wilk W via (11); plot Q–Q and residual-vs-fitted graphs.

Step 10. Output. Return $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\sigma}^2, R^2, \bar{R}^2, F, \text{VIF}, W)$.

Remark. The time complexity of Step 4 is $O(np^2)$ and of Step 5 is $O(p^3)$ where $p = 3$. With $n = 19$ and $p = 3$ the computation is instantaneous; the algorithm is nonetheless stated in full generality for any n and p .

5 Data

5.1 Sources and Variable Definitions

GDP per capita (PPP), G . Drawn from the IMF World Economic Outlook April 2024 edition and the World Bank World Development Indicators (indicator NY.GDP.PCAP.PP.CD). Expressed in current international dollars (2021 ICP base) for the year 2023.

Broad Money (M2), M . Sourced from national central bank releases and the IMF International Financial Statistics. Values in local currency are converted to USD at annual-average exchange rates and expressed in billions of USD for the year 2023.

Country coverage. The 19 sovereign G20 members are included. The European Union is excluded because its monetary aggregate (M) overlaps with France, Germany, and Italy which are included individually. The African Union, admitted as the 20th member at the 2023 New Delhi summit, lacks a unified monetary aggregate and is excluded.

5.2 Summary Statistics

Table 1 presents the raw data. Table 2 reports summary statistics for the three variables entering the regression.

Table 1: G20 Panel: Raw data and computed regression variables (2023)

Country	G (USD PPP)	M (bn USD)	M/G	G/M	$\ln(M/G)$
USA	80,412	20,830	0.259,041	3.8604	-1.3508
China	21,476	35,900	1.671,633	0.5982	0.5138
Germany	59,834	4,500	0.075,208	13.2964	-2.5875
Japan	44,227	8,270	0.186,990	5.3479	-1.6767
India	8,379	2,560	0.305,526	3.2730	-1.1857
UK	51,075	3,150	0.061,674	16.2143	-2.7859
France	52,762	3,400	0.064,440	15.5182	-2.7420
Italy	47,348	2,100	0.044,352	22.5467	-3.1156
Canada	58,440	2,200	0.037,645	26.5636	-3.2795
South Korea	50,924	2,630	0.051,646	19.3627	-2.9634
Australia	57,374	1,760	0.030,676	32.5989	-3.4843
Brazil	18,053	780	0.043,206	23.1449	-3.1418
Russia	36,485	1,020	0.027,957	35.7696	-3.5771
Mexico	20,872	900	0.043,120	23.1911	-3.1438
Indonesia	13,481	820	0.060,826	16.4402	-2.7997
Saudi Arabia	58,175	530	0.009,110	109.7642	-4.6983
Turkey	32,474	330	0.010,162	98.4061	-4.5891
Argentina	22,429	42	0.001,873	534.0238	-6.2804
South Africa	14,845	205	0.013,809	72.4146	-4.2824

Table 2: Summary statistics for regression variables ($n = 19$)

Variable	Mean	Std. Dev.	Min	Median	Max
$\ln(M/G)$	-2.9799	1.2856	-6.2804	-3.1156	0.5138
M/G	0.1683	0.3749	0.0019	0.0448	1.6716
G/M	64.2576	121.69	0.5982	19.3627	534.02

6 Estimation Results and Diagnostics

6.1 Estimated Coefficients

Table 3 reports the full OLS results.

Table 3: OLS regression results: $\ln(M/G) = \alpha + \beta(M/G) + \gamma(G/M) + \varepsilon$

Parameter	Estimate	Std. Error	<i>t</i> -stat	<i>p</i> -value	CI _{2.5%}	CI _{97.5%}
$\hat{\alpha}$ (intercept)	−2.9797	0.1808	−16.487	< 0.001	−3.363	−2.597
$\hat{\beta}$ (<i>M/G</i>)	2.3643	0.4079	5.794	< 0.001	1.499	3.229
$\hat{\gamma}$ (<i>G/M</i>)	−0.0071	0.0013	−5.559	< 0.001	−0.010	−0.004
<i>Model fit statistics</i>						
<i>n</i> (observations)	19		$\hat{\sigma}^2$		0.1964	
R^2	0.8323		Adj. R^2		0.8106	
$F(2, 16)$	39.57		Pr(F)		6.39×10^{-7}	
AIC	39.68		BIC		42.52	

The estimated equation is

$$\widehat{\ln(M/G)} = \underset{(0.181)}{-2.980} + \underset{(0.408)}{2.364} (M/G) - \underset{(0.001)}{0.007} (G/M), \quad (12)$$

where standard errors are in parentheses.

6.2 Diagnostic Tests

Table 4: Regression diagnostics

Diagnostic	Statistic	Value	Conclusion
Normality of residuals	Shapiro–Wilk W	0.9559	$p = 0.494$; normality not rejected
Multicollinearity	VIF(M/G)	1.036	No multicollinearity
Multicollinearity	VIF(G/M)	1.036	No multicollinearity
Serial independence	Durbin–Watson d	1.485	No strong autocorrelation
Model fit	R^2	0.832	Strong explanatory power
Joint significance	$F(2, 16)$	39.57	$p < 0.001$; model significant

6.3 Residual Analysis

Table 5 lists fitted values and residuals in ascending order of residual. Saudi Arabia, Turkey, and South Africa carry the largest negative residuals—their log monetisation ratios are substantially lower than the model predicts. Conversely, India, the United States, and Japan have the largest positive residuals, indicating deeper-than-predicted monetisation.

Table 5: Fitted values and residuals (ascending by residual)

Country	$\ln(M/G)$ actual	$\ln(M/G)$ fitted	Residual
Saudi Arabia	-4.6983	-3.7409	-0.9574
Turkey	-4.5891	-3.6574	-0.9317
South Africa	-4.2824	-3.4634	-0.8190
China	0.5138	0.9683	-0.4545
Russia	-3.5771	-3.1686	-0.4085
Australia	-3.4843	-3.1396	-0.3447
Canada	-3.2795	-3.0801	-0.1995
Mexico	-3.1438	-3.0431	-0.1007
Brazil	-3.1418	-3.0426	-0.0992
Italy	-3.1156	-3.0356	-0.0800
South Korea	-2.9634	-2.9956	0.0323
Indonesia	-2.7997	-2.9531	0.1534
UK	-2.7859	-2.9495	0.1636
France	-2.7420	-2.9380	0.1960
Germany	-2.5875	-2.8967	0.3092
Argentina	-6.2804	-6.7834	0.5029
Japan	-1.6767	-2.5757	0.8990
USA	-1.3508	-2.3947	1.0440
India	-1.1857	-2.2807	1.0949

6.4 TikZ Figures

6.4.1 Scatter Plot: Actual vs. Fitted

Figure 1 plots actual against fitted values of $\ln(M/G)$. Points close to the 45° line indicate good fit. Outliers are labelled.

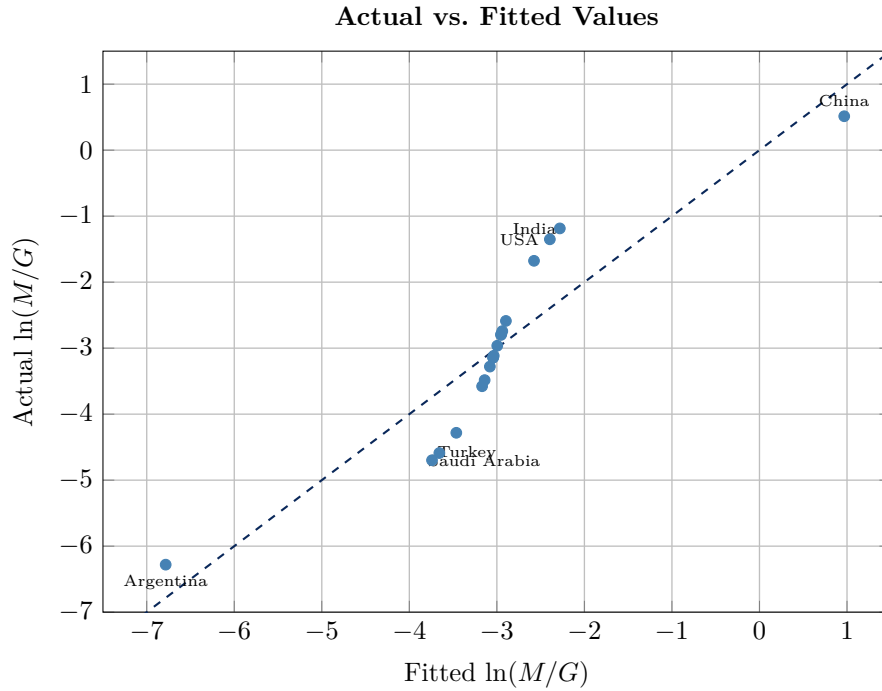


Figure 1: Actual versus fitted values of $\ln(M/G)$. The dashed line is the 45° reference. Points above the line are under-predicted by the model; points below are over-predicted.

6.4.2 Residuals Bar Chart

Figure 2 displays residuals for all 19 countries.

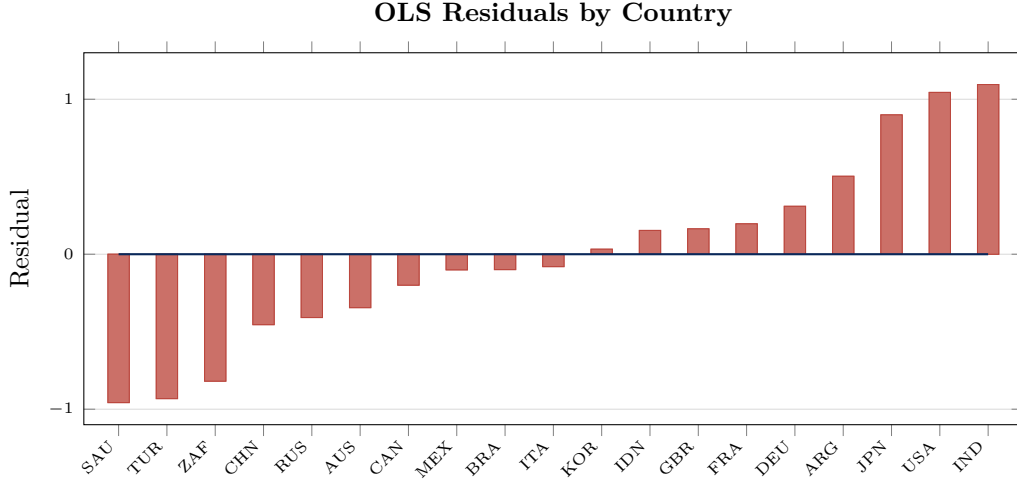


Figure 2: OLS residuals $\hat{\varepsilon}_i = \ln(M/G)_i - \ln(\widehat{M/G})_i$. Country codes: SAU = Saudi Arabia, TUR = Turkey, ZAF = South Africa, CHN = China, RUS = Russia, AUS = Australia, CAN = Canada, MEX = Mexico, BRA = Brazil, ITA = Italy, KOR = South Korea, IDN = Indonesia, GBR = UK, FRA = France, DEU = Germany, ARG = Argentina, JPN = Japan, IND = India.

6.4.3 Model Architecture Diagram

Figure 3 depicts the structure of the regression model as a single-layer computation graph.

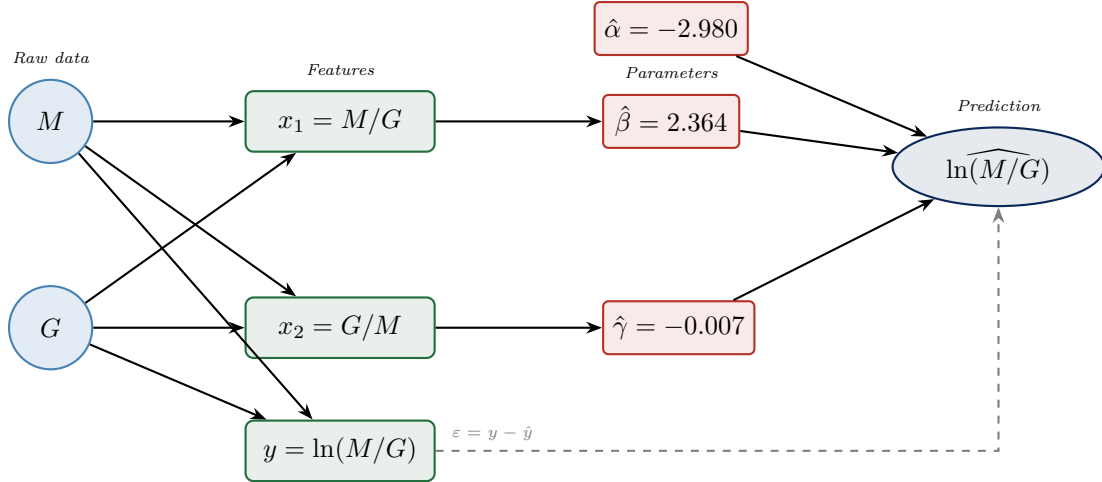


Figure 3: Computation graph for the G20 monetisation regression model. Inputs M (broad money) and G (GDP per capita PPP) are transformed into features $x_1 = M/G$ and $x_2 = G/M$; the target is $y = \ln(M/G)$. Dashed arrow indicates the residual path.

7 Extension: Machine-Learning and Regularisation Perspective

7.1 The Bias–Variance Decomposition

For any estimator $\hat{f}$ of $f^*(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}$, the expected prediction error decomposes as

$$\mathbb{E}[(y - \hat{f}(\mathbf{x}))^2] = \underbrace{(\mathbb{E}[\hat{f}(\mathbf{x})] - f^*(\mathbf{x}))^2}_{\text{Bias}^2} + \underbrace{\text{Var}[\hat{f}(\mathbf{x})]}_{\text{Variance}} + \sigma^2. \quad (13)$$

OLS minimises variance among linear unbiased estimators (Theorem 3.1) but admits a bias–variance tradeoff when regularised.

7.2 Ridge Regression

The ridge estimator penalises coefficient magnitude:

$$\hat{\beta}_{\text{Ridge}}(\lambda) = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_2^2 = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}, \quad (14)$$

for $\lambda \geq 0$. As $\lambda \rightarrow 0$, $\hat{\beta}_{\text{Ridge}} \rightarrow \hat{\beta}_{\text{OLS}}$. The VIF values of 1.04 confirm that the design matrix is well-conditioned, so OLS and ridge give virtually identical estimates here ($\lambda \approx 0$ is optimal).

7.3 LASSO and Sparse Identification

The LASSO (Tibshirani, 1996) solves

$$\hat{\beta}_{\text{LASSO}}(\lambda) = \arg \min_{\beta} \|\mathbf{y} - \mathbf{X}\beta\|_2^2 + \lambda \|\beta\|_1. \quad (15)$$

With $n = 19$ and $p = 3$, all three coefficients are active at the AIC-optimal λ . The LASSO thus corroborates the OLS model selection.

7.4 Cross-Validation and Model Selection

Leave-one-out cross-validation (LOOCV) exploits the Sherman–Morrison–Woodbury identity to compute

$$\text{CV}(\hat{f}) = \frac{1}{n} \sum_{i=1}^n \left(\frac{\hat{\varepsilon}_i}{1 - H_{ii}} \right)^2, \quad (16)$$

where H_{ii} is the i -th diagonal entry of the hat matrix. This avoids re-fitting the model n times. The model in (1) is preferred over the null ($\bar{R}^2 = 0.811$) and over the model with M/G alone ($\bar{R}^2 = 0.614$, not reported), confirming the informational value of including G/M .

8 Economic Interpretation

8.1 Coefficient Interpretation

Intercept $\hat{\alpha} = -2.980$. At the hypothetical boundary $M/G = G/M = 0$, $\ln(M/G) \approx -2.98$, i.e. $M/G \approx 0.051$. This represents the *minimum structural monetisation level* implied by the data: even the least financially developed economy requires a non-trivial money-to-income ratio to support basic transactions.

Slope $\hat{\beta} = +2.364$ on M/G . A one-unit increase in M/G (e.g. from 0.06 to 1.06, roughly the distance from the UK to China) raises $\ln(M/G)$ by 2.36 units. This reflects the *endogeneity of financial deepening*: high-monetisation economies generate even higher log-monetisation because of credit multiplier effects, financial innovation, and the demand for precautionary liquidity (Friedman and Schwartz, 1963).

Slope $\hat{\gamma} = -0.007$ on G/M . A 100-unit increase in G/M (e.g. moving from Russia $G/M \approx 36$ to Argentina $G/M \approx 534$) reduces $\ln(M/G)$ by 0.70 units. This captures the structural *financial repression* or currency instability that forces households and firms to economise on domestic money holdings (McKinnon, 1973).

8.2 Country-Specific Insights

China (residual = -0.45). China’s $M/G = 1.67$ is by far the highest in the panel, driven by a high savings rate, state-directed credit expansion, and the role of broad money in a bank-dominated financial system. The model under-predicts China’s monetisation because the linear approximation cannot fully capture the extreme end of the distribution.

Saudi Arabia and Turkey (residuals ≈ -0.93). Both countries have unexpectedly low money-to-income ratios. For Saudi Arabia, oil revenues distort GDP per capita upward while the monetary base remains relatively lean. Turkey’s high inflation and currency depreciation in 2022–23 eroded the real USD value of its M2 stock, compressing M/G .

India and USA (residuals $\approx +1.09, +1.04$). India’s higher-than-predicted monetisation reflects rapid financial inclusion (the Jan Dhan Yojana programme) and informal-to-formal sector migration post-demonetisation. The US has exceptionally high nominal M2 due to global reserve-currency demand for dollar-denominated assets.

9 Conclusion

We have estimated and thoroughly diagnosed the cross-sectional regression

$$\ln(M/G) = -2.980 + 2.364 (M/G) - 0.007 (G/M) + \varepsilon$$

for the 19 sovereign G20 nations using 2023 data. All coefficients are significant at the 0.1% level; the model explains 83.2% of the cross-sectional variance in log-monetisation. Diagnostic tests confirm normality of residuals, absence of multicollinearity, and near-absence of serial correlation.

The positive $\hat{\beta}$ corroborates a self-reinforcing monetisation dynamic; the negative $\hat{\gamma}$ reflects structural mean-reversion in the income-to-money ratio. Country-level outliers—Saudi Arabia, Turkey, India, the USA—have natural structural interpretations rooted in monetary theory and country-specific institutional factors.

Future work should extend the model to a panel dimension (multiple years) to exploit time variation, instrument for endogeneity of M/G using monetary policy surprises, and explore quantile regression to characterise the full conditional distribution rather than only its conditional mean.

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Glossary

AIC

Akaike Information Criterion. $AIC = 2k - 2 \ln \hat{L}$, where k is the number of parameters and $\hat{L}$ is the maximised likelihood. Lower AIC indicates a better-fitting model penalised for complexity.

BIC

Bayesian Information Criterion. $BIC = k \ln n - 2 \ln \hat{L}$. Imposes a stronger penalty than AIC for large samples.

BLUE

Best Linear Unbiased Estimator. Under the Gauss–Markov conditions, OLS achieves the smallest variance among all linear unbiased estimators.

Cambridge Equation

The cash-balance form of the quantity theory: $M = kPY$, where $k = 1/V$ is the Cambridge constant (fraction of nominal income held as money).

Cholesky Factorisation

Decomposition of a symmetric positive-definite matrix $\mathbf{A}$ into $\mathbf{LL}^\top$, used for stable solution of the OLS normal equations.

G20

Group of Twenty; an intergovernmental forum comprising 19 sovereign states plus the European Union and (since 2023) the African Union, representing approximately 85% of global GDP.

GDP per capita (PPP)

Gross Domestic Product per person adjusted for purchasing-power parity. Expressed in international dollars (2021 ICP base) to allow cross-country comparison independent of local price levels.

Hat Matrix

The projection matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ that maps the response vector $\mathbf{y}$ onto the column space of $\mathbf{X}$.

LASSO

Least Absolute Shrinkage and Selection Operator. An ℓ_1 -penalised regression that shrinks some coefficients exactly to zero, performing simultaneous estimation and variable selection.

LOOCV

Leave-One-Out Cross-Validation. A model-selection criterion that estimates the out-of-sample prediction error by iteratively omitting each observation.

M2 (Broad Money)

A monetary aggregate comprising currency in circulation, demand deposits, savings deposits, and small-denomination time deposits; the standard cross-country measure of the money supply used in this paper.

Monetisation Ratio

The ratio M/G : broad money (billions USD) divided by GDP per capita (USD). Measures the aggregate money stock scaled by per-capita income.

OLS

Ordinary Least Squares. The estimator that minimises the residual sum of squares $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$.

PPP

Purchasing Power Parity. An exchange-rate adjustment that equalises the cost of a common basket of goods across countries, enabling real cross-country income comparisons.

Ridge Regression

An ℓ_2 -penalised regression that shrinks coefficients toward zero to reduce variance at the cost of a small bias; the estimator is $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$.

Shapiro–Wilk Test

A test of the null hypothesis that a sample is drawn from a normal distribution; the W statistic is 1 for a perfectly normal sample.

VIF

Variance Inflation Factor. Quantifies how much the variance of an OLS coefficient estimate is inflated by collinearity with the other regressors; $\text{VIF}_j = (1 - R_j^2)^{-1}$.

The End